

PRACTICAL-1

CODE:

```
GNU nano 7.2 command_execution.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove the newline character added by fgets()
    command[strcspn(command, "\n")] = '\0';

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0)
    {
        // Child process
        printf("\n--- Child Process ---\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());
        printf("Executing command: %s\n\n", command);

        execl("/bin/sh", "sh", "-c", command, NULL);

        // Executes only if execl() fails
        printf("Command execution failed!\n");
    }
}
```

OUTPUT:

1)ls

```
Enter a Linux command: ls

--- Parent Process ---
Parent PID : 771
Child PID  : 772

--- Child Process ---
Child PID : 772
Parent PID: 771
Executing command: ls

command_execution  command_execution.c

Child process has finished execution.
```

2)uname

```

Enter a Linux command: uname

--- Parent Process ---
Parent PID : 798
Child PID  : 801

--- Child Process ---
Child PID : 801
Parent PID: 798
Executing command: uname

Linux

Child process has finished execution.

```

3) lscpu

```

Enter a Linux command: lscpu

--- Parent Process ---
Parent PID : 885
Child PID  : 886

--- Child Process ---
Child PID : 886
Parent PID: 885
Executing command: lscpu

Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i5-1334U
CPU family:             6
Model:                  186
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):               1
Stepping:               3
BogoMIPS:               4992.00
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
                        cp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
                        sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
                        bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflush
                        opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
                        clear serialize ibt flush_l1d arch_capabilities

```

4) lsblk

```

Enter a Linux command: lsblk

--- Parent Process ---
Parent PID : 818
Child PID  : 819

--- Child Process ---
Child PID : 819
Parent PID: 818
Executing command: lsblk

NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda         8:0    0 356.9M  1 disk
sdb         8:16   0 159.4M  1 disk
sdc         8:32   0      2G  0 disk [SWAP]
sdd         8:48   0      1T  0 disk /mnt/wslg/distro
/

```

Child process has finished execution.

5) ps

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-01$ ./command_execution
Enter a Linux command: ps
```

```
--- Parent Process ---
```

```
Parent PID : 821
```

```
Child PID : 822
```

```
--- Child Process ---
```

```
Child PID : 822
```

```
Parent PID: 821
```

```
Executing command: ps
```

```
PID TTY          TIME CMD
305 pts/0        00:00:00 bash
821 pts/0        00:00:00 command_executi
822 pts/0        00:00:00 sh
823 pts/0        00:00:00 ps
```

```
Child process has finished execution
```

6)top

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-01$ ./command_execution
Enter a Linux command: top
```

```
--- Parent Process ---
```

```
Parent PID : 833
```

```
Child PID : 834
```

```
--- Child Process ---
```

```
Child PID : 834
```

```
Parent PID: 833
```

```
Executing command: top
```

```
top - 03:42:26 up 35 min, 1 user, load average: 0.07, 0.06, 0.01
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7762.6 total, 7132.5 free, 523.8 used, 256.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7238.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21644	13124	9796	S	0.0	0.2	0:00.47	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2124	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50364	16300	15212	S	0.0	0.2	0:00.19	systemd-journal
97	root	20	0	25404	6792	5152	S	0.0	0.1	0:00.34	systemd-udev
157	systemd+	20	0	21460	13184	10928	S	0.0	0.2	0:00.09	systemd-resolve
167	systemd+	20	0	91028	7948	6980	S	0.0	0.1	0:00.08	systemd-timesyn
176	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.00	cron
177	message+	20	0	9592	5340	4652	S	0.0	0.1	0:00.09	dbus-daemon